DMFR-YOLO: Defect Morphology Encoding and Feature Reassembly for Fabric Defect Detection

Kangzhao Tan[#], Jiarui Zhang[#], Junjie Yu, Heng Wu, Jing Lin[*], and Chunhua He[*]

***Abstract*—Circular knitting machines are widely used in continuous fabric production. Needle or yarn abnormalities can cause repeated defects along the fabric before they are detected, resulting in material waste and economic losses. To enable in-process defect detection and intervention, we develop a closed-loop defect detection system with edge inference and wired machine-stop control, and propose a lightweight detector termed DMFR-YOLO. DMFR-YOLO introduces defect morphology encoding (DME) into the backbone to explicitly model horizontally elongated defects, vertically elongated defects, and localized hole-like defects. In the feature pyramid, content-adaptive feature reassembly (CAFR) replaces fixed upsampling with content-adaptive reconstruction, improving the preservation of defect boundaries and weak responses during multi-scale feature fusion. Evaluated on a self-collected fabric defect dataset from a real-world circular knitting machine production line, DMFR-YOLO achieves 66.3% AP and 96.9% AP50, improving the YOLOv8n baseline by 7.0 and 2.8 percentage points, respectively, with only 3.39 M parameters. The detector is deployed as an INT8 RKNN model on an RK3588 edge device. The resulting system couples real-time defect detection, automatic local stop control, operator review, and cloud-based quality analysis, providing a practical solution for reducing defect propagation during circular-knitting production.**

***Index Terms*—Industrial defect detection, YOLO, deep learning, edge computing, Internet of Things.**

I. Introduction

CIRCULAR knitting machines are important production infrastructure for high-volume weft-knit fabrics used in T-shirts and sportswear [1]–[3]. The International Textile Manufacturers Federation reported approximately 24,500 global shipments of large circular knitting machines in 2025 [4]. Globally, approximately 92 million tonnes of textile waste are generated annually, highlighting the importance of reducing avoidable material losses [5]. Conventional quality checks are often performed after knitting is completed. A persistent needle or yarn abnormality can therefore generate repeated defects along a fabric roll before discovery, accumulating substantial fabric waste and economic losses [6]–[8]. Real-time detection during knitting is needed to identify these defects and enable intervention before further defective fabric is produced.

Hardware studies have demonstrated several approaches to automatic defect detection. A circular-knitting system combines a Basler line-scan camera, encoder-triggered acquisition, and a desktop computer [9]. Other fabric studies deploy inference on Jetson TX2 or Orin Nano platforms [10], [11], while a glass defect system implements a dedicated accelerator on a ZCU102 development board [12]. Customized fabric detection hardware has also been reported, including an intelligent camera based on Cambricon 1H8 [13]. These studies establish the feasibility of local processing, but their hardware configurations address different operating conditions and integration requirements. For circular knitting machines, computing capability must be matched to the detection workload and cost constraints, while camera installation, video transmission, and machine-control interfaces must be adapted to the production equipment. A general-purpose computing platform alone does not provide this application-specific integration, motivating a dedicated system designed around the circular-knitting workflow.

At the algorithm level, YOLO-based detectors provide efficient inference and explicit defect localization [14]. Recent studies have incorporated YOLOv3 into an optical detection system for LCD module defects [15], combined template assistance and attention with YOLOv7 for paper defects [16], and introduced feature prefusion and mask guidance into a Scaled-YOLOv4-style detector for camera-decoration surface defects [17]. Lightweight convolution and structural reparameterization have further supported insulator and rail defect detection on embedded platforms [18], [19]. These works demonstrate the practicality of the YOLO route and improve defect representation or inference efficiency. For circular-knitted fabrics, our focus is to retain this efficiency while explicitly representing the characteristic shapes of defects in repetitive backgrounds.

The defects considered in this work appear as horizontally elongated defects, vertically elongated defects, and localized holes. Standard convolutional feature extraction can learn these patterns implicitly, but it does not explicitly separate their spatial responses. In addition, the nearest-neighbor upsampling used in conventional feature pyramids reconstructs higher-resolution features through fixed replication rather than local content-dependent aggregation. Preserving thin defect boundaries and weak local responses under a limited computational budget therefore motivates morphology-specific feature encoding and adaptive reconstruction during multi-scale fusion.

Based on these hardware and algorithm considerations, we design and implement a dedicated defect detection system for circular knitting machines and propose a lightweight detector, DMFR-YOLO. The system integrates an industrial camera with LED illumination, a Hall sensor, a coaxial video link, and a self-developed RK3588 edge board. Machine rotation activates detection, and a detected defect triggers a local wired stop command and displays the result for operator review. The cloud receives defect and device records for monitoring

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and production-quality analysis without participating in the stop decision [20]. Within the detector, defect morphology encoding (DME) introduces horizontal, vertical, and local branches into the backbone, while content-adaptive feature reassembly (CAFR) replaces fixed top-down upsampling with locally predicted neighborhood aggregation. The hardware integration and compact detector jointly support detection and intervention during knitting.

The main contributions of our work are as follows.

(1) We design and implement a dedicated detection system for circular knitting machines, integrating image acquisition, camera transmission, RK3588 edge inference, and local machine-stop control on application-specific hardware, and demonstrate its deployment on a real production line.
(2) We introduce DME to encode the characteristic morphologies of horizontal defects, vertical defects, and holes through horizontal, vertical, and local branches, providing a lightweight morphology-aware representation for circular-knitted defect detection.
(3) We incorporate CAFR into the feature pyramid to replace fixed upsampling with locally predicted reassembly weights, enabling content-adaptive reconstruction during multi-scale fusion.

II. Hardware and Detection System

A. Hardware Configuration

The system combines image acquisition, machine-state sensing, edge computation, and a wired connection to the knitting-machine controller. Fig. 1 shows the camera assembly and our self-developed RK3588 edge-device hardware. The industrial camera and LED light are integrated in the front assembly (A). We use a VEYE MV-MIPI-SC130M monochrome camera with a SmartSens SC130GS-M global-shutter sensor and a 1280×1024 imaging resolution [21]. A camera-side interface board (B) provides a serializer and a coaxial connector, while the RK3588 edge board (C) includes a deserializer for the camera connection. This arrangement connects the camera mounted near the fabric to the edge device responsible for detection and machine control.

The RK3588 edge board integrates flash memory, power management, and input/output interfaces, together with a 4G module, SIM-card slot, and antenna connector for cloud communication. The Hall sensor provides the machine-rotation signal, and a wired connection to the machine controller carries the stop signal. Image processing and the stop decision remain on the edge device; cloud communication is used for device monitoring and defect records rather than for latency-critical control.

Fig. 2 summarizes the interconnections among the system components. Camera data pass through the camera-side serializer, coaxial link, and edge-side deserializer before reaching the RK3588 processor. The Hall sensor supplies the rotation signal through the input/output interface, which also connects the edge device to the knitting-machine controller for local stop control. The power adapter supplies the camera assembly and edge device. The 4G module provides cloud connectivity, while the machine-control path remains local.

B. Closed-Loop Detection Workflow

Fig. 3 shows the installation of our defect detection system on a circular knitting machine production line. After the operator starts the circular knitting machine, the Hall sensor detects rotation and signals the RK3588 edge device to switch on the LED light and begin detection from the camera video stream. When a defect is detected, the device sends a stop signal to the machine controller through a wired connection to interrupt further defect propagation. The defect image and detection result are displayed locally for operator review and troubleshooting. The operator resolves the problem and restarts the machine, completing the operational cycle.

C. Edge Deployment and Cloud Platform

The detector described in Section III is deployed locally on the RK3588 edge device. To enable on-device inference, the trained model is converted from PyTorch to ONNX and then to an INT8 RKNN model. The trained model is exported with a fixed 640×640 input and converted using RKNN-Toolkit2 [22], with post-training INT8 quantization for efficient NPU execution [23]. An Android application executes the RKNN model on the RK3588 edge device.

The cloud platform receives defect records and device states for centralized monitoring, defect statistics, and production-quality analysis. Its interface supports device-state visualization, remote configuration, and review of uploaded defect images (Fig. 4).

D. Comparison With Existing Defect Detection Hardware

Published defect detection studies use different computing and imaging arrangements. Song et al. deploy EfficientDet on a Jetson TX2 for fabric defect detection [10], while Machado et al. use a Jetson Orin Nano with two RealSense D435i cameras for textile defect detection [11]. In other defect detection applications, Li et al. evaluate an insulator detector on a Jetson TX2 NX [18], Wang et al. employ a Jetson Xavier NX for rail surface defects [19], and Zhao et al. implement glass defect segmentation on a ZCU102 FPGA platform [12]. Table I identifies the task associated with each platform; unrelated object detection applications are not included.

Table II compares camera models explicitly identified in fabric defect detection studies with the camera used in our system. The circular-knitting system of Çelik et al. uses a Basler line-scan camera connected to a desktop computer [9]; it is included as an imaging-system comparison, not as an embedded inference platform. Camera specifications describe the selected products, whereas field of view, illumination, installation, and processing hardware remain application-specific. The reference prices therefore characterize individual components rather than a matched end-to-end system cost.

Our implementation combines a global-shutter camera, a coaxial camera link, local NPU inference, and machine-control interfaces on a self-developed edge board. This integration is tailored to installation on circular knitting machines. The hardware comparison describes deployment choices across tasks; it is not a cross-dataset accuracy benchmark or a measurement of complete-system cost.

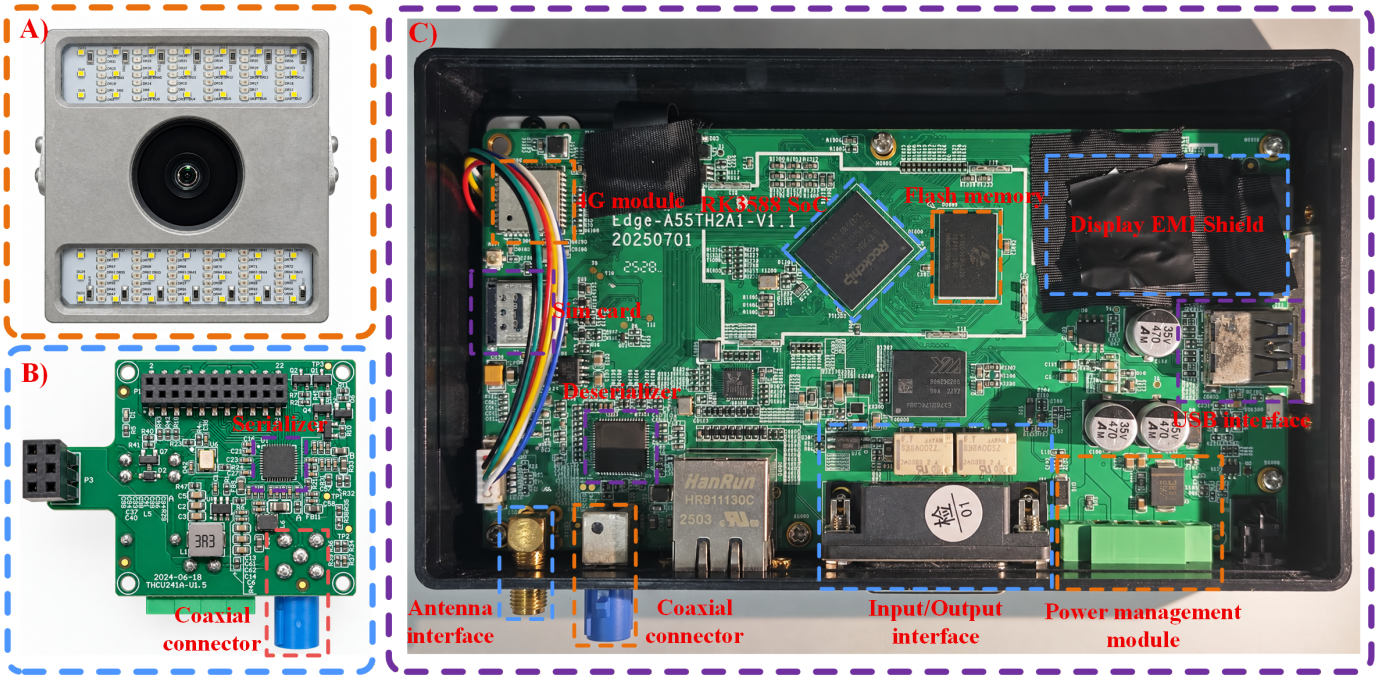


Fig. 1. Hardware of the production-line defect detection system: (A) industrial camera with LED illumination; (B) camera-side interface board with a serializer and coaxial connector; and (C) RK3588 edge board with the labeled communication, memory, camera-interface, input/output, and power-management components.

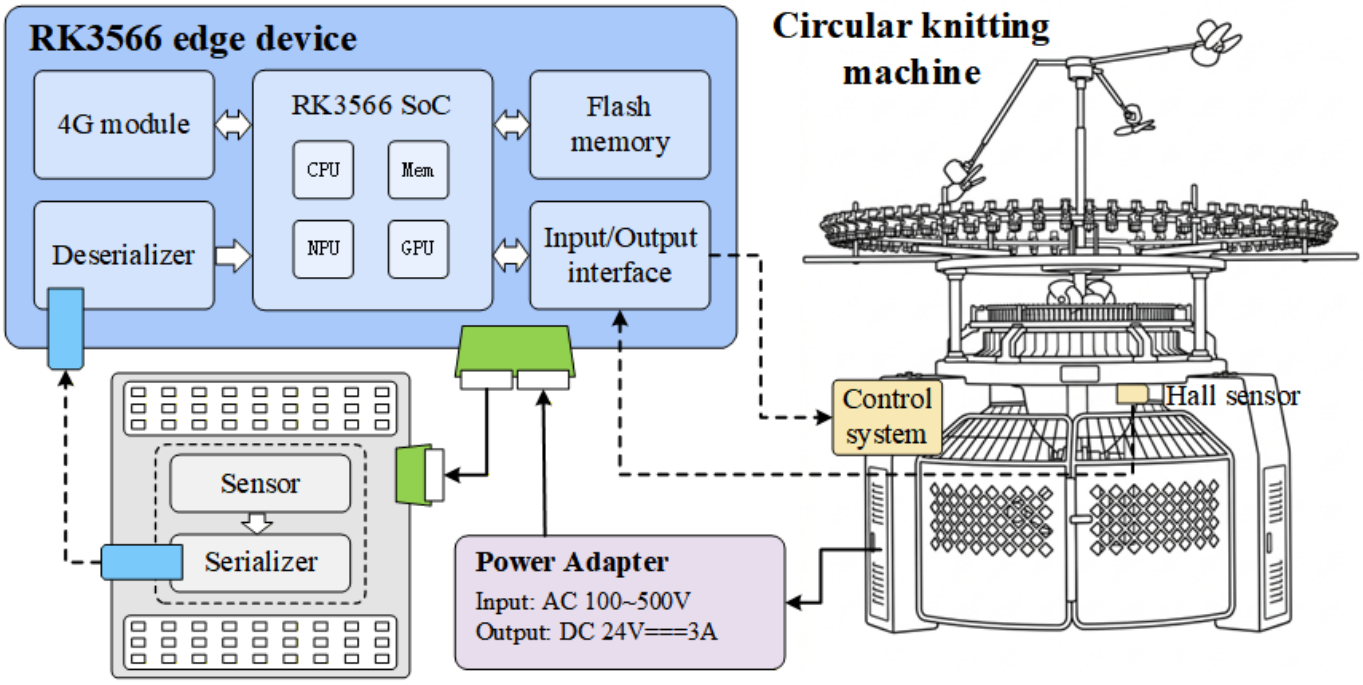


Fig. 2. System component interconnections. The camera stream reaches the RK3588 edge device through the serializer, coaxial link, and deserializer. Wired input/output connections carry the Hall-sensor signal and stop command, while the power adapter supplies the camera and edge hardware.

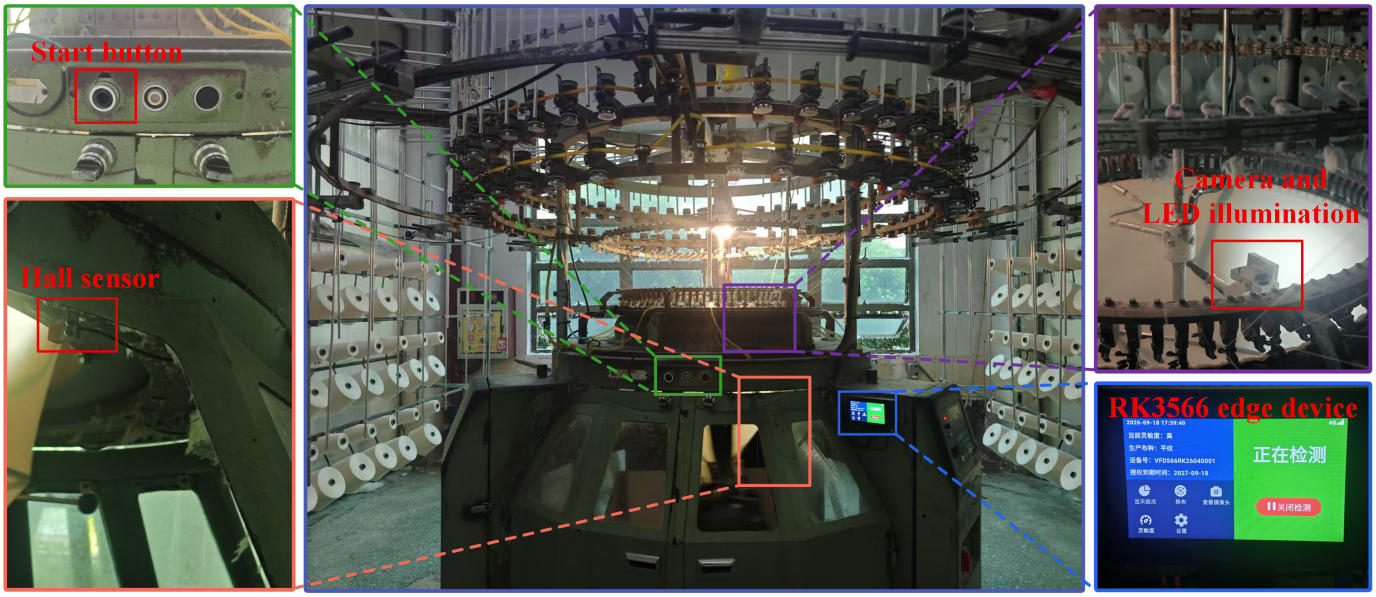


Fig. 3. Actual deployment on the circular knitting machine. The insets highlight the start button, Hall sensor, industrial camera, and RK3588 edge device, respectively.

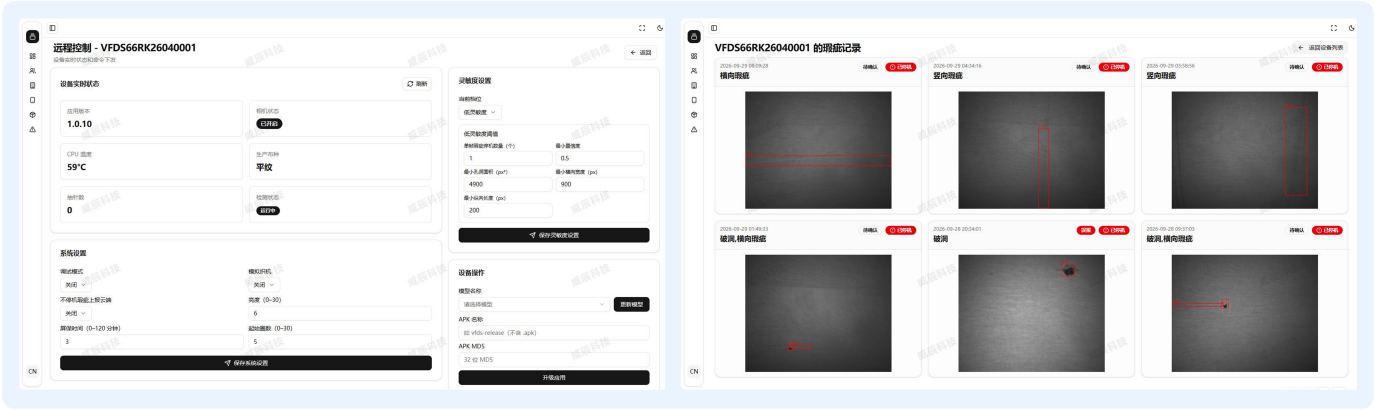


Fig. 4. IoT cloud interface: device-state monitoring and remote configuration (left), and uploaded defect records (right). Cloud functions support centralized supervision and quality analysis.

TABLE I

PROCESSING PLATFORMS IN SELECTED DEFECT DETECTION STUDIES AND OUR SYSTEM. PRICES ARE INDICATIVE COMPONENT REFERENCE PRICES, NOT THE COSTS OF THE COMPLETE SYSTEMS IN THE CITED STUDIES.

Study	Detection task	Processing platform	Memory	Compute capability	Price (USD)
Song et al. [10]	Fabric defects	Jetson TX2	8 GB	1.33 TFLOPS (FP16)	599
Li et al. [18]	Insulator defects	Jetson TX2 NX	4 GB	1.33 TFLOPS (FP16)	249
Wang et al. [19]	Rail surface defects	Jetson Xavier NX	8 GB	21 TOPS (INT8)	399
Machado et al. [11]	Textile defects	Jetson Orin Nano	8 GB	40 TOPS (sparse INT8)	499
Zhao et al. [12]	Glass defects (segmentation)	Xilinx ZCU102 (ZU9EG)	4 GB PS; 0.5 GB PL	Implementation-dependent	3234
Our system	Circular-knitted fabric defects	Self-developed RK3588 edge board	NR	6 TOPS (INT8 NPU)	NR

NR: not reported. Memory and nominal compute ratings describe the listed reference configurations; they do not establish measured model throughput or the operating mode used by each study. FP16 TFLOPS and INT8 TOPS are different quantities [24]–[26]. ZCU102 has separate PS/PL memory, and KAM-Net uses a custom 16-bit fixed-point accelerator rather than a standard INT8 DPU [12], [27]. The RK3588 NPU rating follows the manufacturer specification [28]. Prices refer to modules or development kits: TX2 and Orin Nano use kit reference prices, TX2 NX and Xavier NX use module reference prices, and ZCU102 uses an evaluation-kit price. No commercial-board price is assigned to our self-developed board.

TABLE II
CAMERA CONFIGURATIONS IN SELECTED FABRIC DEFECT DETECTION SYSTEMS. PRICES ARE INDICATIVE REFERENCE PRICES FOR INDIVIDUAL CAMERA UNITS.

Study	Detection task	Camera model	Type and key specifications	Interface	Price (USD)
Çelik et al. [9]	Circular-knitted fabric defects	Basler raL2048-48gm	Monochrome line scan; 2048×1; global-shutter CMOS; 8/12-bit	GigE	1647
Song et al. [10]	Fabric defects	Basler acA2500-14gm	Monochrome area scan; 2592×1944; rolling-shutter CMOS; 12-bit	GigE	629
Machado et al. [11]	Textile defects	Intel RealSense D435i	Active-stereo RGB-D; depth 1280×720; RGB 1920×1080; IMU	USB 3.0	349
Our system	Circular-knitted fabric defects	VEYE MV-MIPI-SC130M (SC130GS-M)	Monochrome area scan; 1280×1024; global shutter; $4\,\mu$m pixels; 1/2.7-inch format	MIPI CSI-2	51

Product specifications follow manufacturer documentation [21], [29]–[31]. Camera prices are per unit, not per imaging system, and do not include lenses, lighting, transmission boards, mounting hardware, or integration. The D435i configuration in Machado et al. uses two cameras.

III. Proposed Method

A. DMFR-YOLO Architecture

DMFR-YOLO is developed based on YOLOv8n, which is selected for its compatibility with RK3588 deployment, compact model size, and modular architecture [32], [33]. The proposed modifications focus on defect morphology encoding in the backbone and feature reconstruction in the feature pyramid.

Fig. 5 shows the overall architecture. DME is introduced after the main backbone stages to enhance the representation of different defect morphologies. The extracted multi-scale features are then passed to the neck for feature fusion [34], [35]. Within the feature pyramid, CAFR replaces two nearest-neighbor upsampling operations with content-adaptive feature reassembly to better preserve defect boundaries and weak responses. The fused features are subsequently used by the multi-scale detection head to predict defect locations and categories.

B. Defect Morphology Encoding

The three defect categories considered in this work present different spatial characteristics. Horizontal and vertical defects usually extend along one dominant direction, whereas hole defects are more localized. To capture these differences, we introduce a defect morphology encoding (DME) module with three parallel branches, as shown in the enlarged view in Fig. 5. The horizontal and vertical branches use 1×7 and 7×1 depthwise convolutions to enlarge the receptive field along the corresponding spatial directions, while the 3×3 branch captures localized defect structures. Depthwise convolution is adopted to model these spatial patterns with low computational overhead before channel-wise feature fusion [36].

For a C2f output $X \in \mathbb{R}^{C \times H \times W}$, the three branch responses are

$$F_h = \mathrm{DW}_{1\times7}(X), \tag{1}$$

$$F_v = \mathrm{DW}_{7\times1}(X), \tag{2}$$

$$F_l = \mathrm{DW}_{3\times3}(X). \tag{3}$$

Here, DW denotes depthwise convolution followed by batch normalization and SiLU activation. All branches preserve the spatial dimensions and produce C output channels. Their concatenation is fused as

$$F_d = \phi\left([F_h, F_v, F_l]\right), \tag{4}$$

where $[\cdot]$ denotes channel concatenation and ϕ is a 1×1 convolution mapping $3C$ channels to C, followed by batch normalization and SiLU. This learnable fusion combines the three types of spatial evidence rather than assigning a fixed branch to a particular defect class. A residual connection [37] gives

$$Y = X + F_d. \tag{5}$$

The residual connection combines the morphology-aware features with the incoming C2f output, allowing DME to enhance the backbone representation without discarding the features learned by the preceding layers. The directional branches are based on asymmetric convolution designs [38].

C. Content-Adaptive Feature Reassembly

In the feature pyramid, nearest-neighbor upsampling enlarges feature maps by directly replicating feature values. This fixed operation does not adapt to the spatial structure of defects. To

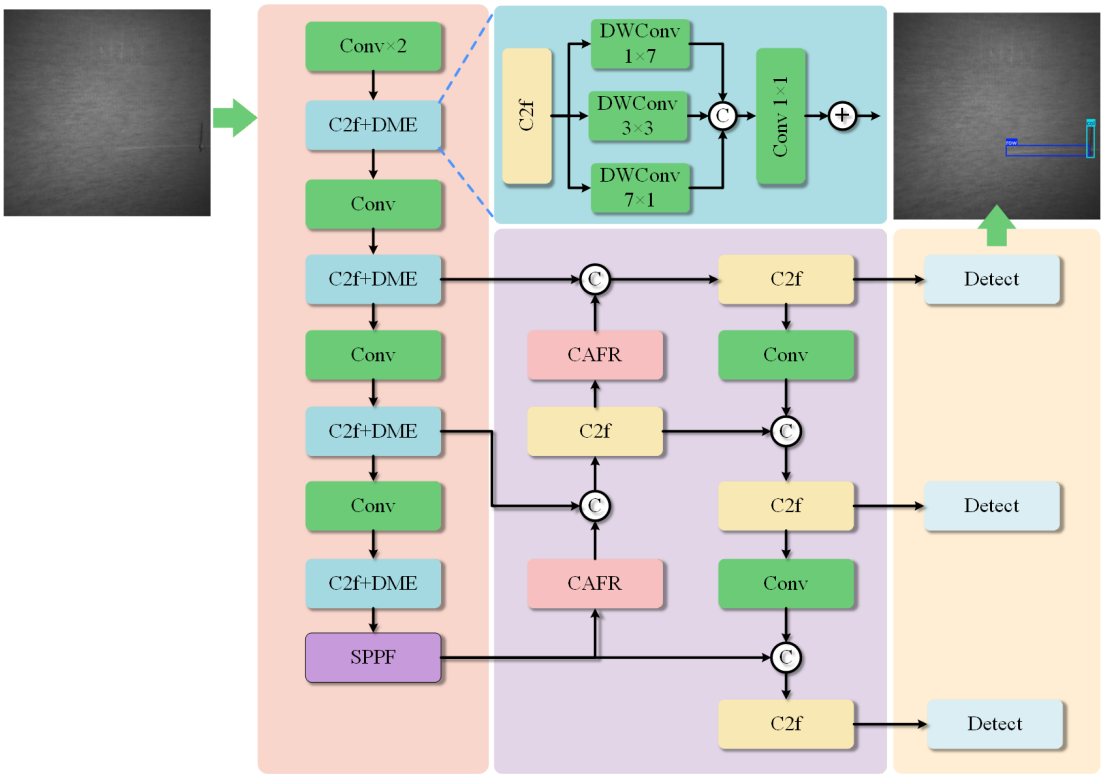


Fig. 5. Architecture of DMFR-YOLO with an enlarged view of DME. DME follows the backbone C2f stages, and CAFR replaces the two top-down upsampling operations. The neck fusion blocks, bottom-up path, and three-scale detection head are retained from YOLOv8n.

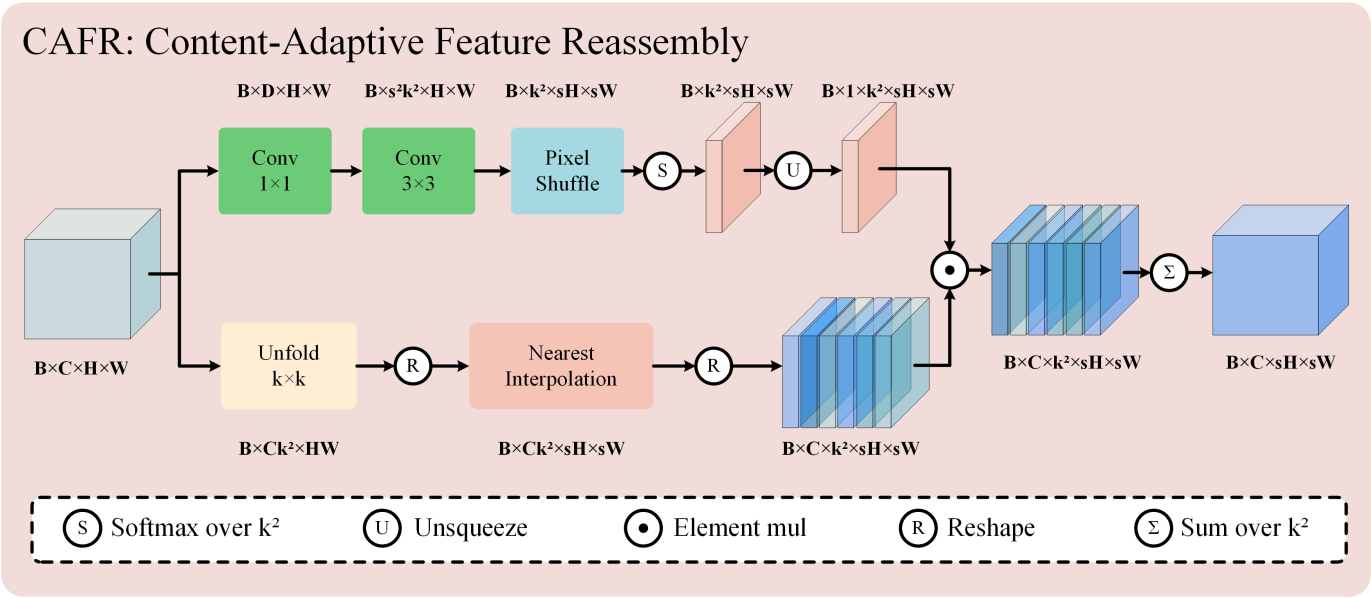


Fig. 6. CAFR structure. The upper path predicts location-specific 5×5 reassembly kernels, and the lower path unfolds local feature patches and replicates them to align with output locations. Weighted aggregation, rather than this alignment operation, produces the final $2\times$ upsampled feature.

address this limitation, we introduce content-adaptive feature reassembly (CAFR), following the reassembly mechanism of Content-Aware ReAssembly of FEatures(CARAFE) [39]. CAFR predicts spatially varying kernels from the input features and uses them to aggregate local neighborhoods (Fig. 6), aiming to better preserve defect boundaries and weak responses during multi-scale fusion.

In DMFR-YOLO, CAFR replaces the upsampling operations at both top-down fusion stages, with an upsampling factor of $s = 2$ and a reassembly kernel size of $k = 5$. As shown in Fig. 6, CAFR consists of two parallel branches for kernel prediction and local feature extraction. Their outputs are subsequently combined through weighted aggregation to generate the upsampled features. The two branches and their subsequent feature reassembly are described as follows:

1) Reassembly Kernel Prediction: Given an input feature

map $X \in \mathbb{R}^{C\times H\times W}$, a 1×1 convolution first reduces its channel dimension:

$$X_c = E_{1\times 1}(X). \tag{6}$$

Here, $X_c \in \mathbb{R}^{D\times H\times W}$, where $D = \max(\lfloor C/4 \rfloor, 16)$ denotes the compressed channel count; $E_{1\times 1}$ includes batch normalization and SiLU activation. A linear 3×3 convolution then predicts the unnormalized reassembly weights:

$$K' = E_{3\times 3}(X_c). \tag{7}$$

The resulting tensor is $K' \in \mathbb{R}^{s^2k^2\times H\times W}$. For $s = 2$ and $k = 5$, this corresponds to 100 weight channels at each input location. Pixel Shuffle rearranges the predicted weights onto the higher-resolution grid, followed by Softmax normalization:

$$K = \mathrm{Softmax}(\mathrm{PS}(K')). \tag{8}$$

Softmax is applied along the k^2 neighborhood dimension. The resulting tensor $K \in \mathbb{R}^{k^2\times sH\times sW}$ provides a normalized reassembly kernel for each output location.

2) Local Feature Extraction: In parallel, the patch branch extracts local neighborhoods from the input feature map using Unfold. The extracted patches are reshaped and upsampled using nearest-neighbor interpolation to match the spatial resolution of the predicted reassembly kernels:

$$\begin{aligned} P' &= \mathrm{Reshape}(\mathrm{Unfold}_{k\times k}(X)), \\ P &= \mathrm{Reshape}(\mathrm{Nearest}_s(P')). \end{aligned} \tag{9}$$

Here, $\mathrm{Unfold}_{k\times k}$ extracts a $k\times k$ neighborhood around each input position, and $\mathrm{Nearest}_s$ denotes nearest-neighbor upsampling with scale factor $s = 2$. The first reshape produces $P' \in \mathbb{R}^{Ck^2\times H\times W}$; the final reshape gives the aligned patch tensor $P \in \mathbb{R}^{C\times k^2\times sH\times sW}$, where k^2 indexes the positions within each local neighborhood. Nearest-neighbor interpolation is used only for neighborhood alignment, not to generate the final output features.

3) Feature Reassembly: The predicted kernels and the aligned local features are combined through element-wise multiplication and summation along the neighborhood dimension:

$$Y = \sum_{u=1}^{k^2} K_u \odot P_u. \tag{10}$$

Here, $K_u \in \mathbb{R}^{sH\times sW}$ and $P_u \in \mathbb{R}^{C\times sH\times sW}$ denote the weight map and corresponding local feature map for the u-th neighborhood position, respectively. The symbol $\odot$ denotes element-wise multiplication, with the weights broadcast across feature channels, producing an output feature map $Y \in \mathbb{R}^{C\times sH\times sW}$.

By predicting location-specific reassembly weights, CAFR adaptively aggregates local features during upsampling, helping preserve defect-related information in multi-scale feature fusion.

IV. Experimental Setup

A. Dataset

Images were collected during operation of a real circular-knitting production line. A global-shutter industrial camera captures 1280×1024 frames approximately 10 cm from the fabric, with a field of view of approximately 16×14 cm. We selected images with visible defects and manually annotated their bounding boxes. The dataset includes three defect categories: horizontal defects (*row*), vertical defects (*col*), and holes (*hole*).

The dataset contains 403 images with 468 annotated defect instances, divided into training, validation, and test sets of 282, 40, and 81 images, respectively, as summarized in Table III. The validation set is used for checkpoint selection, while performance is reported on the test set containing 91 labeled defects.

TABLE III
IMAGE-LEVEL SPLIT AND DEFECT-INSTANCE COUNTS IN THE REAL PRODUCTION-LINE DATASET.

Split	Images	Instances	Row	Col	Hole
Train	282	329	174	75	80
Validation	40	48	24	13	11
Test	81	91	60	12	19

B. Implementation Details

Table IV summarizes the training and deployment settings. DMFR-YOLO and YOLOv8n are initialized using the pre-trained yolov8n.pt weights, while newly added layers use default initialization. Other comparison models use the available pretrained checkpoints in their respective implementations. The remaining Ultralytics training settings follow the framework defaults.

The comparisons include lightweight YOLO models [32], [40], DETR [41], Deformable DETR [42], Faster R-CNN [43], and an R18 model from the RT-DETRv2 implementation [44], denoted RT-DETR-R18. All models are evaluated on the same test set using their respective frameworks and evaluation settings.

C. Evaluation Metrics

We report COCO average precision (AP) across ten IoU thresholds and AP at IoU 0.50 (AP50) [45]. For a predicted

TABLE IV
IMPLEMENTATION PROTOCOL. TRAINING SETTINGS DESCRIBE THE ULTRALYTICS EXPERIMENTS; DEPLOYMENT SETTINGS DESCRIBE THE INSTALLED EDGE DEVICE.

Category	Parameter	Specification
1. Training setup	Hardware	Intel i7-13700KF; NVIDIA RTX 3090 (24 GB)
	Software	Ubuntu 24.04, Python 3.10.20, PyTorch 2.4.1+cu124, and Ultralytics 8.4.140
2. Deployment setup	Hardware	Rockchip RK3588
	System	Android 13
	Quantization	INT8
3. Data protocol	Input resolution	Fixed at 640×640
4. Training dynamics	Batch	16
	Maximum epochs	300
	Early-stopping patience	80 epochs

TABLE V
DETECTION RESULTS ON THE TEST SET. BOLD AND UNDERLINED VALUES INDICATE THE HIGHEST AND SECOND-HIGHEST REPORTED SCORES, RESPECTIVELY.

Model	Params (M)	FLOPs (G)	AP50 (%)	AP (%)
Lightweight YOLO Detectors				
YOLOv5n	2.50	7.10	93.1	58.4
YOLO11n	2.58	6.40	95.0	59.6
YOLO12n	2.56	7.30	94.3	60.9
YOLO26n	2.38	5.30	96.5	61.4
Transformer Detectors				
RT-DETR-R18	20.09	60.72	95.2	**67.7**
DETR-R50	41.28	77.25	96.4	53.6
Deformable DETR-R50	39.82	158.66	<u>96.8</u>	64.3
Two-Stage Detector				
Faster R-CNN-R50	41.13	181.45	96.1	64.7
Baseline				
YOLOv8n	3.01	8.10	94.1	59.3
Ours				
DMFR-YOLO	3.39	8.90	**96.9**	<u>66.3</u>

box B_p and a ground-truth box B_g, intersection over union is $\mathrm{IoU}(B_p, B_g) = |B_p \cap B_g|/|B_p \cup B_g|$. Let $\mathrm{AP}_c(t)$ denote the average precision for class c at IoU threshold t, and let N_c denote the number of defect categories. The metrics are defined as follows:

$$\mathrm{AP} = \frac{1}{10N_c} \sum_{c=1}^{N_c} \sum_{t \in T} \mathrm{AP}_c(t), \tag{11}$$

$$\mathrm{AP}_{50} = \frac{1}{N_c} \sum_{c=1}^{N_c} \mathrm{AP}_c(0.50), \tag{12}$$

where $T = \{0.50, 0.55, \ldots, 0.95\}$ and $N_c = 3$ in this study. All AP values are reported as percentages. AP therefore evaluates localization over increasingly strict overlap requirements, whereas AP50 uses a single, less restrictive threshold. Differences between reported AP values are expressed in percentage points.

Model parameters and FLOPs are measured using the respective frameworks with an input resolution of 640 × 640.

V. RESULTS AND DISCUSSION

A. Comparison With Other Detectors

Table V presents the detection results on the test set. DMFR-YOLO achieves 66.3% AP and 96.9% AP50, improving the YOLOv8n baseline by 7.0 and 2.8 percentage points, respectively. These improvements are achieved with only 0.38 M additional parameters and approximately 0.8 G additional FLOPs. The larger improvement in AP indicates better detection performance under stricter IoU thresholds.

Among the evaluated lightweight YOLO detectors, DMFR-YOLO achieves the highest AP. Although RT-DETR-R18 attains a slightly higher AP of 67.7%, it requires 20.09 M parameters and 60.72 G FLOPs, compared with 3.39 M parameters and 8.90 G FLOPs for DMFR-YOLO. Fig. 7 further illustrates the favorable balance between detection accuracy and model complexity achieved by DMFR-YOLO.

Fig. 8 compares the detection results of different models on seven selected test images, highlighting differences in detection accuracy and bounding-box localization.

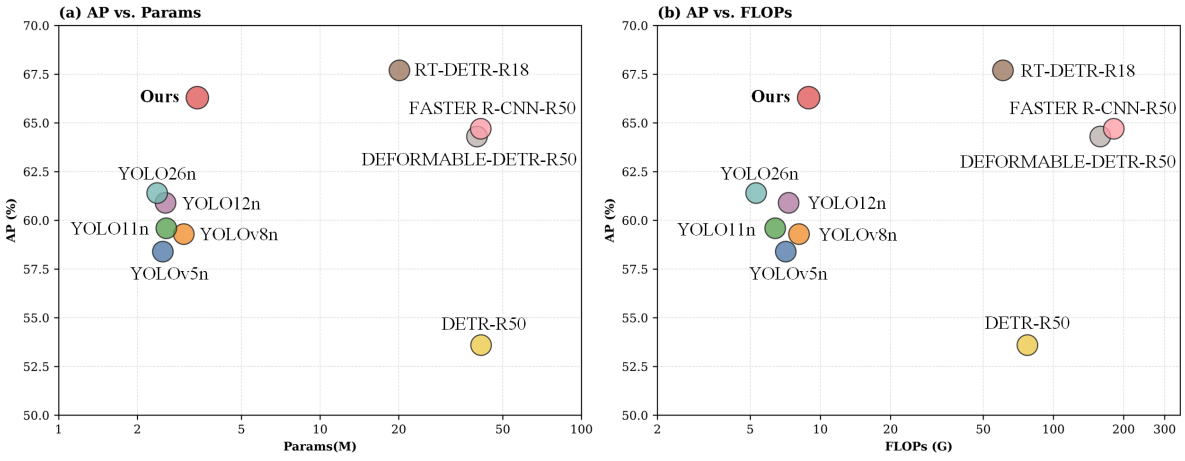


Fig. 7. AP versus Params (left) and FLOPs (right). Both horizontal axes are logarithmic. Ours denotes DMFR-YOLO.

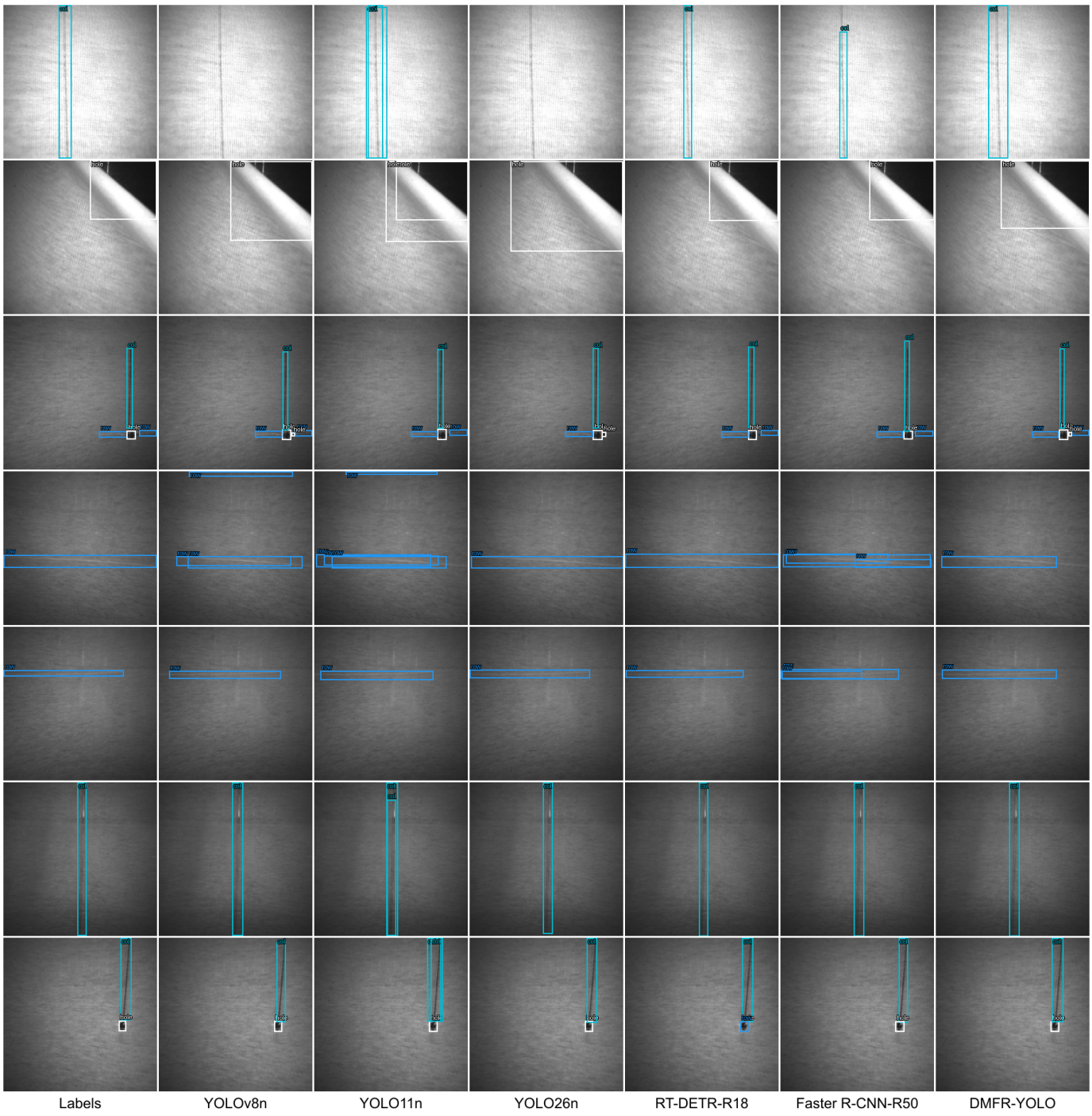


Fig. 8. Qualitative comparison of detection results on selected test images. From left to right, the columns show ground-truth annotations, YOLOv8n, YOLO11n, YOLO26n, RT-DETR-R18, Faster R-CNN-R50, and DMFR-YOLO (Ours).

B. Component Ablation

Table VI evaluates the individual and combined effects of DME and CAFR on detection performance. Introducing DME increases AP from 59.3% to 64.1%, while CAFR alone improves AP to 60.7%. When both modules are incorporated, DMFR-YOLO achieves 66.3% AP and 96.9% AP50, outperforming either individual modification. These results demonstrate the complementary effects of the two modules. DME enhances the representation of defect morphologies in the backbone, while CAFR adaptively reconstructs these features during multi-scale fusion. Their combination further improves the overall detection performance.

Fig. 9 compares the feature visualizations of YOLOv8n and DMFR-YOLO. Compared with YOLOv8n, DMFR-YOLO exhibits more concentrated feature responses around elongated defects and holes, with activation regions more closely aligned with the defect shapes. These observations suggest that DMFR-YOLO provides a more focused representation of defect regions.

TABLE VI
COMPONENT ABLATION. PARAMETERS ARE IN MILLIONS, FLOPS IN BILLIONS, AND AP50/AP IN PERCENT.

Variant	DME	CAFR	Params	FLOPs	AP50	AP
YOLOv8n	–	–	3.01	8.1	94.1	59.3
+ DME	✓	–	3.28	8.8	94.2	64.1
+ CAFR	–	✓	3.11	8.2	94.9	60.7
DMFR-YOLO	✓	✓	3.39	8.9	**96.9**	**66.3**

VI. CONCLUSION

This paper presents DMFR-YOLO and a defect detection system for circular knitting machines. DMFR-YOLO introduces defect morphology encoding (DME) to capture the spatial characteristics of horizontal defects, vertical defects, and holes. Content-adaptive feature reassembly (CAFR) improves feature reconstruction during multi-scale fusion. Experiments on the production line dataset show that DMFR-YOLO achieves 66.3% AP and 96.9% AP50, outperforming YOLOv8n by 7.0 and 2.8 percentage points, respectively, with 3.39 M parameters and 8.9 G FLOPs. The ablation results further demonstrate the complementary effects of DME and CAFR.

The proposed detector is deployed on an RK3588 edge device and integrated with machine monitoring and wired stop control. The system enables automatic defect detection and timely intervention without relying on cloud communication. Detection results are displayed locally for operators, while the cloud platform supports equipment monitoring and production quality analysis. By integrating defect detection with machine control and quality management, this work provides a complete and practical solution for fabric defect detection in circular knitting production. Future work will explore the application of DMFR-YOLO to different knitted fabrics with varying texture patterns and defect characteristics to evaluate its generalization across production conditions.

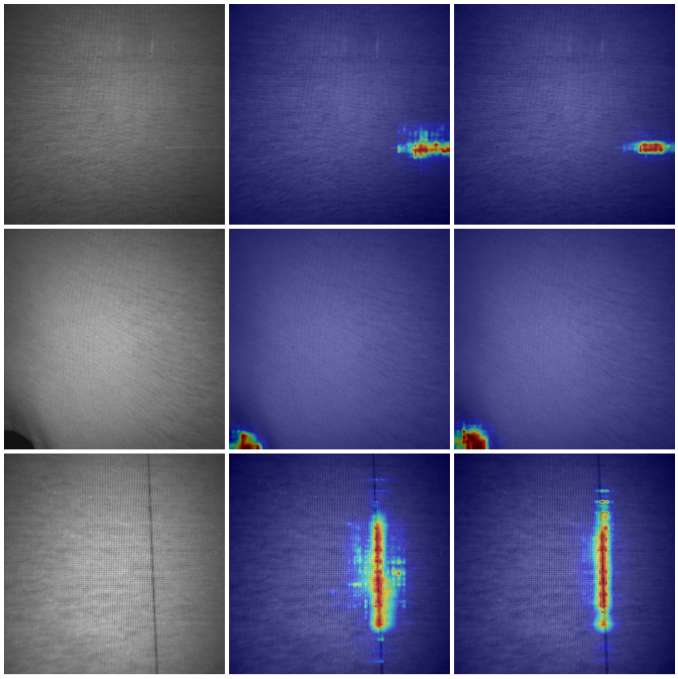


Fig. 9. Feature visualization of YOLOv8n and DMFR-YOLO. From left to right: original image (Origin), YOLOv8n, and DMFR-YOLO.

REFERENCES

[1] Cotton Incorporated, "Knit basics," CottonWorks, [Online]. Available: https://cottonworks.com/learning-hub/knitting/knit-basics/. Accessed: Oct. 8, 2026.
[2] The Woolmark Company, "Circular knit technology," [Online]. Available: https://www.woolmark.com/industry/product-development/processing-innovations/circular-knitting-technology/. Accessed: Oct. 8, 2026.
[3] Cotton Incorporated, "Single and double knits," CottonWorks, [Online]. Available: https://cottonworks.com/learning-hub/knitting/single-and-double-knits/. Accessed: Oct. 8, 2026.
[4] International Textile Manufacturers Federation, "Global textile machinery shipments shrunk in 2025, except for spinning," Jul. 2026, [Online]. Available: https://www.itmf.org/images/dl/press-releases/2026/2026-07-24-ITMSS-48-2025.pdf. Accessed: Oct. 10, 2026.
[5] United Nations Environment Programme, "Transforming the textile industry: Learning to manage waste, better," United Nations Environment Programme, Tech. Rep., Oct. 2025, [Online]. Available: https://wedocs.unep.org/handle/20.500.11822/48707. Accessed: Oct. 14, 2026.
[6] S. Takeuchi, K. Nishioka, H. Uematsu, and S. Tanoue, "Research into development of the defect detection system for knitted fabric produced by the circular knitting machines by image analysis," *Journal of Textile Engineering*, vol. 64, no. 2, pp. 45–49, 2018.
[7] M. M. K. Akter, U. N. Haq, M. M. Islam, and M. A. Uddin, "Textile-apparel manufacturing and material waste management in the circular economy: A conceptual model to achieve sustainable development goal (SDG) 12 for bangladesh," *Cleaner Environmental Systems*, vol. 4, p. 100070, 2022.
[8] Z. Xiang, Y. Shen, M. Ma, and M. Qian, "HookNet: Efficient multiscale context aggregation for high-accuracy detection of fabric defects," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023, art. no. 5016311.
[9] H. İ. Çelik, L. C. Dülger, B. Öztaş, M. Kertmen, and E. Gültekin, "A novel industrial application of CNN approach: Real time fabric inspection and defect classification on circular knitting machine," *Tekstil ve Konfeksiyon*, vol. 32, no. 4, pp. 344–352, 2022.
[10] S. Song, J. Jing, Y. Huang, and M. Shi, "EfficientDet for fabric defect detection based on edge computing," *J. Eng. Fibers Fabrics*, vol. 16, pp. 1–13, 2021.
[11] R. Machado, L. A. M. Barros, V. Vieira, F. D. da Silva, H. Costa, and V. Carvalho, "Textile defect detection using artificial intelligence and computer vision—a preliminary deep learning approach," *Electronics*, vol. 14, no. 18, p. 3692, 2025.
[12] B. Zhao, Y. Wang, H. Zhang, J. Liu, and J. Zhang, "KAM-Net: Kilobyte-scale ultralightweight attention-based network for glass defect detection with algorithm/hardware co-design," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–10, 2025, art. no. 5041310.
[13] Z. Zhu, G. Han, G. Jia, and L. Shu, "Modified DenseNet for automatic fabric defect detection with edge computing for minimizing latency," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9623–9636, 2020.
[14] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779–788.
[15] H. Zhu, J. Huang, H. Liu, Q. Zhou, J. Zhu, and B. Li, "Deep-learning-enabled automatic optical inspection for module-level defects in LCD," *IEEE Internet Things J.*, vol. 9, no. 2, pp. 1122–1135, 2022.
[16] X. Li, H. Yan, K. Cui, Z. Li, R. Liu, G. Lu, K. C. Hsieh, X. Liu, and C. Hon, "A novel hybrid YOLO approach for precise paper defect detection with a dual-layer template and an attention mechanism," *IEEE Sensors J.*, vol. 24, no. 7, pp. 11 651–11 669, 2024.
[17] H. Wang, Y. Zhao, F. Zhang, G. Gui, Q. Luo, C. Yang, and W. Gui, "A feature prefusion and mask-guided network for camera decoration defect detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–10, 2024, art. no. 5039510.
[18] D. Li, Y. Lu, Q. Gao, X. Li, X. Yu, and Y. Song, "LiteYOLO-ID: A lightweight object detection network for insulator defect detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–12, 2024, art. no. 5023812.
[19] Y. Wang, H. Yu, B. Guo, H. Shi, and Z. Yu, "Research on real-time detection system of rail surface defects based on deep learning," *IEEE Sensors J.*, vol. 24, no. 13, pp. 21 157–21 167, 2024.
[20] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision

and challenges," *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, 2016.
[21] VEYE Imaging, "MV-MIPI-SC130M product specifications," [Online]. Available: https://www.veye.cc/en/product/mv-mipi-sc130m/. Accessed: Oct. 14, 2026.
[22] Rockchip, "Rknn toolkit2," [Online]. Available: https://github.com/airockchip/rknn-toolkit2, 2026, accessed: 2026-09-27.
[23] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 2704–2713.
[24] NVIDIA, "Jetson TX2 series technical specifications," [Online]. Available: https://www.nvidia.com/en-sg/autonomous-machines/embedded-systems/jetson-tx2/. Accessed: Oct. 14, 2026.
[25] ——, "Introducing Jetson Xavier NX, the world's smallest AI supercomputer," 2019, [Online]. Available: https://developer.nvidia.com/blog/?p=15799. Accessed: Oct. 14, 2026.
[26] ——, "NVIDIA Jetson Orin Nano sets new standard for entry-level edge AI and robotics with 80x performance leap," 2022, [Online]. Available: https://nvidianews.nvidia.com/news/nvidia-jetson-orin-nano-sets-new-standard-for-entry-level-edge-ai-and-robotics-with-80x-performance-leap. Accessed: Oct. 14, 2026.
[27] AMD, "ZCU102 evaluation board user guide," UG1182, [Online]. Available: https://docs.amd.com/api/khub/documents/rt6tROKcAceWoROXFPMY5w/content. Accessed: Oct. 14, 2026.
[28] Rockchip Electronics, "RK3588 brief datasheet," [Online]. Available: https://www.rock-chips.com/uploads/pdf/2022.8.26/191/RK3588%20Brief%20Datasheet.pdf. Accessed: Oct. 14, 2026.
[29] Basler, "Basler racer raL2048-48gm product specifications," [Online]. Available: https://www.baslerweb.com/en-us/shop/ral2048-48gm/. Accessed: Oct. 14, 2026.
[30] ——, "Basler ace Classic acA2500-14gm product specifications," [Online]. Available: https://www.baslerweb.com/en-us/shop/aca2500-14gm/. Accessed: Oct. 14, 2026.
[31] Intel, "Intel RealSense depth camera D435i specifications," [Online]. Available: https://www.intel.com/content/www/us/en/products/sku/190004/intel-realsense-depth-camera-d435i/specifications.html. Accessed: Oct. 14, 2026.
[32] Ultralytics, "Ultralytics yolo," [Online]. Available: https://github.com/ultralytics/ultralytics, 2023, accessed: 2026-09-24.
[33] Rockchip, "Rknn model zoo," [Online]. Available: https://github.com/airockchip/rknn_model_zoo, 2026, accessed: 2026-09-27.
[34] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2117–2125.
[35] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8759–8768.
[36] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "MobileNets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.
[37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
[38] X. Ding, Y. Guo, G. Ding, and J. Han, "Acnet: Strengthening the kernel skeletons for powerful cnn via asymmetric convolution blocks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 1911–1920.
[39] J. Wang, K. Chen, R. Xu, Z. Liu, C. C. Loy, and D. Lin, "Carafe: Content-aware reassembly of features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 3007–3016.
[40] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," *arXiv preprint arXiv:2502.12524*, 2025.
[41] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.
[42] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Representations*, 2021.
[43] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," in *Adv. Neural Inf. Process. Syst.*, vol. 28, 2015.
[44] W. Lv, Y. Zhao, Q. Chang, K. Huang, G. Wang, and Y. Liu, "Rt-detrv2: Improved baseline with bag-of-freebies for real-time detection transformer," *arXiv preprint arXiv:2407.17140*, 2024.
[45] T.-Y. Lin, M. Maire, S. Belongie *et al.*, "Microsoft coco: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.

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